Dutch Disease or Dutch Blessing? Shale Gas Shock and Education Decisions

(preliminary draft)

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Contents

Introduction

Motivation

- **Dutch Disease** refers to the reallocation of resources from capital-intensive sectors to booming resource-based sectors, which hinders the growth of the former and increases the region's or country's dependence on natural resources.
    - Consistent with classical trade theory (Rybczynski Theorem), existing empirical research has primarily focused on developing economies.
    - also related to **educational decisions**, since capital-intensive sectors typically require more education than resource-based sectors
- The massive **shale gas discoveries** in the 2000s, centered around the Barnett basin (TX) and the Marcellus basin (PA), provide a compelling setting for testing this theory in a developed economy context.

Goals

Empirical Side

- Based on Sun and Abraham (2021), provide *inter-state* comparisons on the impact of the shale gas shock on three key variables: **household income**, **high school attendance**, and **college decisions**.
- It investigates heterogeneity in these effects (mainly across different income levels).

Econometric Side

- assess the heterogeneity in a more data-driven way(Athey and Imbens 2016), extending R-learners(Wager and Athey 2018) to a panel or a repeated cross-section framework

Literature Review

- Positive Effects of Shale Gas Shock **on Income**
  - Higher per capita income in shale counties(Weinstein and Partridge 2011)
- Negative Effects of Shale Gas Shock **on Education**
  - Decline in high school and college attainment in West Virginia, Montana, and North Dakota(Rickman, Wang, and Winters 2017)
  - Increased dropout rates among immigrants in Texas(Carpenter, Anderson, and Dudensing 2019)
- Limitations in Prior Studies and our Contribution
  - Not much discussions of **heterogeneous treatment effects**
  - Mostly intra-state county-level comparisons, inter-state comparisons rely on SCMs(Abadie, Diamond, and Hainmueller 2010)

Empirical Strategy

IPUMS ACS Data

- IPUMS ACS - 1yr from 2000 to 2010
- Variables of Interest : Household Income, Education Attainment, School Attendance, etc.
- excluded five states (Hawaii, Alaska, Puerto Rico, Colorado, and Wyoming)
- Covariates are mostly balanced, but shale states tend to have lower income (documented widely in prev. research)

	Age	Observations	Family Income			Education (Years)		
			Mean	Std. Dev.	Count	Mean	Std. Dev.	Count
Non-shale	[5, 10]	1,308,465.00	72,838.58	75,229.40	1,298,525.00	0.63	1.31	1,308,465.00
	[16, 18]	716,276.00	76,209.93	75,288.26	684,920.00	10.60	1.32	716,276.00
	[18, 24]	1,396,758.00	59,542.08	63,236.34	1,287,939.00	12.37	2.02	1,396,758.00
	total	17,029,516.00	71,707.67	72,064.18	16,680,293.00	11.14	5.28	16,429,863.00
Shale	[5, 10]	301,477.00	62,547.86	64,017.72	299,038.00	0.60	1.25	301,477.00
	[16, 18]	162,607.00	67,489.58	66,903.25	155,650.00	10.48	1.40	162,607.00
	[18, 24]	315,031.00	52,141.72	55,290.96	291,304.00	12.22	2.00	315,031.00
	total	3,794,579.00	63,292.86	62,940.60	3,709,749.00	10.74	5.21	3,656,064.00

Table 1: Summary Statistics(Mean and SD weighted)

Methodology - Sun and Abraham (2021) DiD

Year	State
2005	Texas
2006	Arkansas
2006	Oklahoma
2008	Alabama
2008	Louisiana
2008	Pennsylvania
2008	West Virginia

Table 2: Shale Gas Extraction Beginnings

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	t_1	t_2	t_3	t_4	t_5
Texas(g_1)	o	o	o	o	o
Arkansas(g_2)	x	o	o	o	o
Oklahoma(g_2)	x	o	o	o	o
Alabama(g_4)	x	x	x	o	o
Louisiana(g_4)	x	x	x	o	o
Pennsylvania(g_4)	x	x	x	o	o
West Virginia(g_4)	x	x	x	o	o
Other States(g_∞)	x	x	x	x	x

Table 3: Staggered Adoption

Methodology - Sun and Abraham (2021) DiD

- De Chaisemartin and d'Haultfoeuille (2020) shows that if (a) the treatment is staggeredly rolled out and (b) the TEs are heterogeneous across g and t, TWFE is biased.
- Assumption 3(Treatment Effect Homogeneity) of Sun and Abraham (2021)
  - $E[Y_{i,e+s} - Y^{\infty}_{i,e+s} \mid E_i = e] = \alpha_s \quad \forall e \in \{1, \ldots, T, \infty\}$
  - i.e., the dynamic treatment effects(event-study parameters) are equal across different groups.
- Regression Units: household(household income), individuals(highschool attendance and college enrollment)
  - ATE: $Y_{ist} = \alpha_s + \delta_t + \sum_{k=-5}^{5} \beta_k \times \text{Year}_k(i, s, t) + \epsilon_{ist}$
  - HTE: $Y_{iqst} = \alpha_{sq} + \delta_t + \sum_{k=-5}^{5} \beta_k \times \text{Year}_k(i, s, q, t) + \epsilon_{isqt}$

Results

Main Result: Income

- Household income increased by 2.5% overall in the third year after the shale boom.
- 18% increase in 1Q and 5% increase in 2Q.

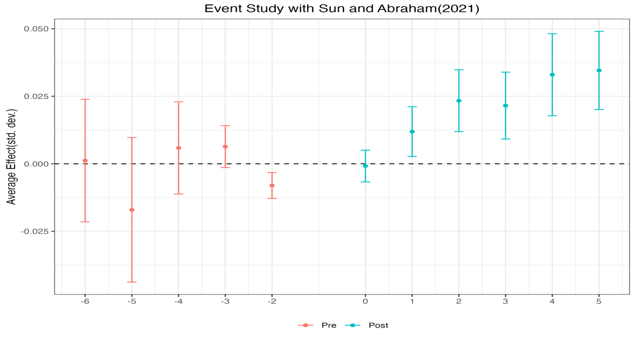

Figure 1: Overall Income

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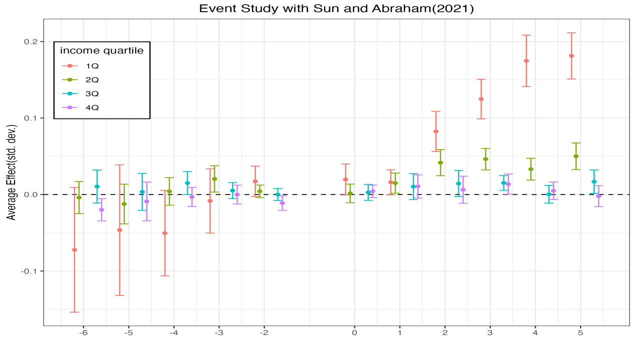

Figure 1: Income by Quartile

Main Result: Highschool Attendance

- Highschool attendance decreased by 0.8% overall in the third year after the shale boom.
- increase in 1Q households(18%)

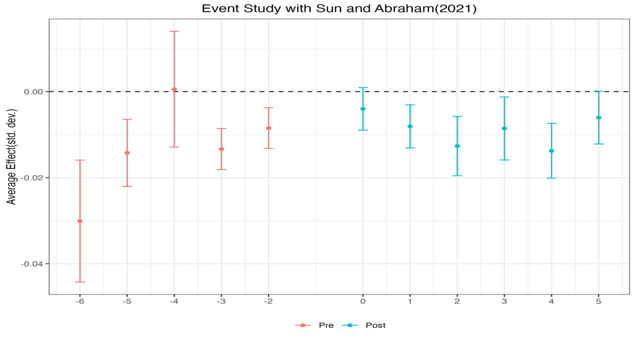

Figure 2: Highschool Attendance (Overall)

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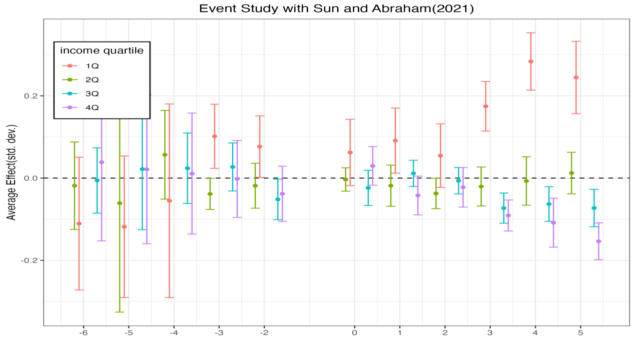

Figure 2: Highschool Attendance by Income Quartile

Main Result: College Enrollment

- College enrollment increased by 7.5%p overall in the third year after the shale boom.
- increase in 1Q households(4.8%p)

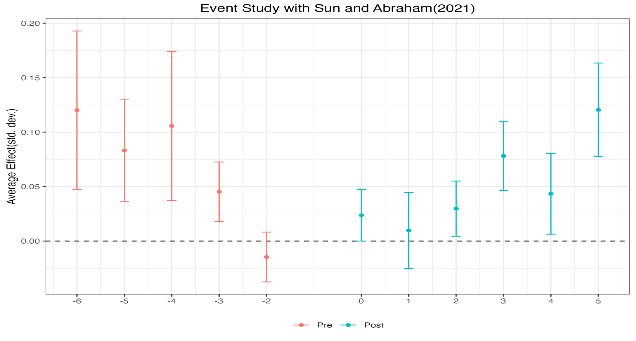

Figure 3: College Enrollment (Overall)

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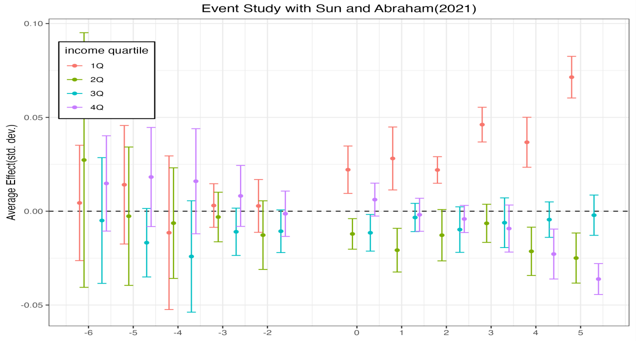

Figure 3: College Share by Income Quartile

Validity of the Identification Assumptions

Parallel Trends

- Some evidence of pre-trends and non-linearity:
  - Household income (1Q): slight upward pre-trend, but post-treatment increase is larger and steeper.
  - hs attendance and college enrollment: significant pre-trend, but disappears when controlling for income $\Rightarrow$ supports *conditional* parallel trends.
- Solutions: Athey et al. (2025), Athey et al. (2021), Arkhangelsky et al. (2021)

Compositional Changes

- Concern: selective immigration from non-shale to shale states may bias estimates upward.
- Disproof: shale states had more outflow than inflow post-treatment. ▸ Details

Conclusion and Next Steps

Conclusion

Shale gas boom increased income and college enrollment, especially for low-income households, casting a doubt on the conventional wisdom of the "Dutch Disease". But I haven't figured out why.

- Previous studies reporting an increase in income and a decrease in education are consistent with the traditional trade theory and the *Dutch disease* hypothesis, but we cannot explain why income and education are moving together
- **Next Step(1):** Investigating which students stayed in school, and which students dropped out, will guide us on the characteristics that lead to different decisions.

Next Steps(2)

Choosing income quartiles for HTE analysis is an arbitrary setup(Athey and Imbens 2016). Why do we divide the sample by their household income, and why four?

- How can we incorporate ML methods such as causal forest to this research?
  - Wang, Martinez, and Hahn (2024) proposes the following method to estimate CATT in panel data settings:
  - Outcome Model

$$Y_{it} = \alpha\mu(X_i, T = t) + \beta_S \nu(X_i, S_{it}, T = t) + \epsilon_{it}$$

    where S is the length to exposure (Train XBART forests separately for μ and ν)
  - CATT:

$$\tau_t(X_i, S) \equiv \mathbb{E}[Y_{it}|X_i, S_{it} = S] - \mathbb{E}[Y_{it}|X_i, S_{it} = 0]$$

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- Drawback: a type of T-learner, which has known problems identifying CATTs(e.g. regularization bias) Example

Next Steps(2)

The ideal solution in the metalearner literature is to use the **R-learner**(Künzel et al. 2019, Chernozhukov et al. 2022): $\sqrt{n}$-consistent and follows the CLT under reasonable limits

$$\begin{aligned} Y_i &= \mu_(0)(X_i) + W_i \tau(X_i) + \varepsilon_i & (1) \\ &= m(X_i) + (W_i - e(X_i))\tau(X_i) + \varepsilon_i & (2) \end{aligned}$$

where $\mathbb{E}[\varepsilon_i | X_i, W_i] = 0$ and

$$\begin{aligned} e(x) &= \mathbb{E}[W_i | X_i = x] & (3) \\ m(x) &= \mathbb{E}[Y_i | X_i = x] = \mu_{(0)}(x) + e(x)\tau(x) & (4) \end{aligned}$$

which allows the following *residuals-on-residuals*

$$Y_i - m(X_i) = \big(W_i - e(X_i)\big)\tau(X_i) + \varepsilon_i$$

Next Steps(2)

DGP for the Panel Case(with mostly the same assumptions as in Sun and Abraham (2021))

$$Y_{it}(S) \perp < D_{i1}, D_{i2}, \dots, D_{iT} > |X_i \qquad \forall S = 0, 1, \dots T \tag{5}$$

$$Y_{it} = \sum_{s=1}^{T} \mathbb{I}(S_{it} = s)\tau(X_i, s) + g_0(X_i, \delta_t) + \varepsilon_{it} \tag{6}$$

$$D_{it} = m_o(X_i) + \nu_{it} \tag{7}$$

where ε and ν satisfies $\mathbb{E}[\varepsilon|D, X] = 0$ and $\mathbb{E}[\nu|X] = 0$.

Equation (6) implies that for untreated individuals and periods,

$$Y_{it} - Y_{it'} = g_0(X_i, \delta_t) - g_0(X_i, \delta'_t) \qquad \forall t \neq t'$$

Thank You:)

Appendix

Appendix

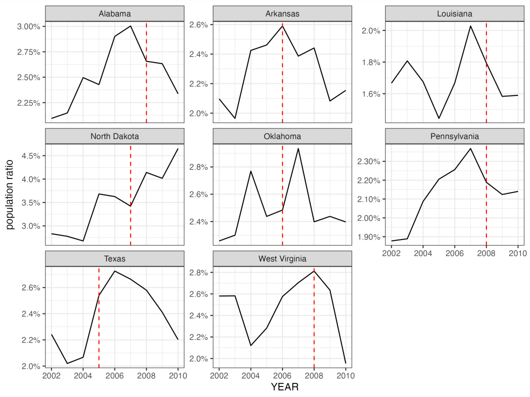

Figure 4: Migration Rate Before and After the Shale Gas Shock

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Appendix

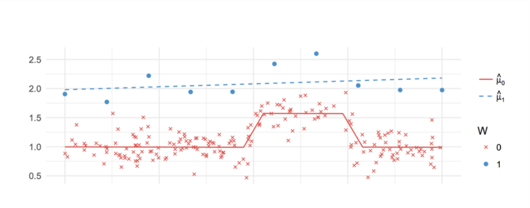

Figure 5: Example from Künzel et al. (2019) on T-learners

return

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